The Telegraph Model of Gene Expression: A Triangular Cross-Validation of Stochastic Simulation

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Abstract

Gene expression—the process by which a gene's DNA is read out to make messenger RNA (mRNA), the molecule that carries its instructions—is a stochastic process: even genetically identical cells in the same environment produce very different numbers of mRNA molecules. The telegraph model explains the main source of this variability. It treats a gene as a two-state (ON/OFF) switch that transcribes mRNA only while active. We study the telegraph model in three independent ways and check that all three agree: we (i) generate stochastic trajectories with the Gillespie stochastic simulation algorithm (SSA); (ii) derive and solve a closed system of ordinary differential equations (ODEs) for the first and second moments; and (iii) draw independent samples directly from the analytic steady-state distribution, using the classic Beta–Poisson law together with an explicit Bernoulli gene state. The main contribution is the cross-validation that links the three methods. The code is available at https://github.com/zakhar-guzii/telegraph-model-gene-expression.

1 Introduction

A surprising fact of molecular biology is that gene expression is variable. Two cells with the same DNA, grown together in the same environment, can have different numbers of copies of the same mRNA or protein [6, 7]. This difference between cells is called phenotypic noise. It happens naturally because of random molecular interactions. Experimentally this variability separates into two parts [6]: *extrinsic* noise, caused by differences between cells—their size, cell-cycle stage, or the pool of available polymerases—and *intrinsic* noise, the randomness of the transcription and degradation events themselves, which persists even between two identical copies of a gene inside the *same* cell. The telegraph model studied here describes the intrinsic component. Inside a cell the key players—a single copy of the gene, a handful of transcription factors (regulatory proteins that bind DNA and control gene activity), a few dozen mRNA molecules—are present in such small numbers that the discreteness and randomness of individual chemical events cannot be averaged away. The promoter (the DNA region that controls when a gene is switched on) switches between an active and an inactive state at random times. While active, it produces mRNA molecules through transcription (the synthesis of mRNA from the gene's DNA) in random bursts. Each mRNA molecule is then degraded—broken down and removed—at a random time. As a result, the mRNA count is a discrete-valued stochastic process, not a deterministic quantity.

The telegraph model is the simplest framework that reproduces this behaviour. It idealises the gene as a two-state device—like a telegraph key that is either pressed (ON) or released (OFF)—that produces mRNA only while ON [8]. Despite its simplicity, it reproduces the bursty, *over-dispersed* mRNA distributions seen in single-molecule experiments, which count individual mRNAs inside single cells [5], and it can be analysed exactly [4]. A convenient way to measure this spread is the *Fano factor* $F = \sigma^2/\mu$ [9], the ratio of the variance to the mean of the mRNA count. Pure Poisson production and decay, in which every event is independent, has variance equal to mean, giving the *Poisson baseline* $F = 1$. "Over-dispersed" means $F > 1$: the random switching of the gene adds extra variability on top of that baseline.

We treat the telegraph model as a case study in how a stochastic process can be described from several mathematically distinct viewpoints. We implement three approaches that look unrelated:

1. a stochastic simulation (the SSA), which answers "what does a single cell actually do?";
2. a system of moment ODEs, which answers "what does the average cell do, and how spread out is the population?";
3. a steady-state sampler, which answers "what does the population look like after a very long time?".

Each approach is correct by itself. But only by comparing them can we be sure that all three are built correctly and that our understanding of the model is right.

2 Model Formulation

The state of the system at time t is the pair

$$\big(G(t),\, R(t)\big), \qquad G(t) \in \{0,1\}, \quad R(t) \in \mathbb{N}, \tag{1}$$

where the binary variable G records whether the gene is OFF ($G = 0$) or ON ($G = 1$), and the integer R counts the number of mRNA molecules present. Four basic reactions define the dynamics of the system. Each reaction has a reaction intensity a_j, which is the rate at which it happens at any given moment:

Reaction	**Description**	**Effect**	**Reaction intensity**
R_1	Gene activation	$G : 0 \to 1$	$a_1 = k_{\text{on}}\,(1 - G)$
R_2	Gene inactivation	$G : 1 \to 0$	$a_2 = k_{\text{off}}\,G$
R_3	RNA synthesis	$R \to R + 1$	$a_3 = k_{\text{syn}}\,G$
R_4	RNA degradation	$R \to R - 1$	$a_4 = k_{\text{deg}}\,R$

All four rate constants are positive. The switching rates $k_{\text{on}}, k_{\text{off}}$ set how quickly the promoter turns on and off, and their ratio fixes the long-run fraction of time the gene is active; the factor G in a_3 shuts transcription off entirely when the gene is OFF; and k_{deg} is a per-molecule decay rate, so the total degradation intensity $k_{\text{deg}}R$ grows with the amount of mRNA present. Together the rates $(k_{\text{on}}, k_{\text{off}}, k_{\text{syn}}, k_{\text{deg}})$ and an initial condition (g_0, r_0) at time t_0 fully determine the law of the process.

Burst size. One derived quantity is used repeatedly below and follows in a line. Once the gene is ON, the only reaction that can switch it off fires at the constant rate k_{off}, so the duration of an active period is exponential with mean $1/k_{\text{off}}$; during that period transcription fires at rate k_{syn}. The mean number of transcripts made per active period—the *burst size* B—is therefore

$$B = k_{\text{syn}}\,\mathbb{E}[\text{ON duration}] = \frac{k_{\text{syn}}}{k_{\text{off}}}. \tag{2}$$

Bursts of this kind are observed directly in single cells [10, 5], and the exact consequences of promoter switching for the mRNA distribution, including the limits recovered in Section 4.1, are worked out in [11].

Because each waiting time is exponential and depends only on the current state, $(G(t), R(t))$ is a continuous-time Markov chain (CTMC): a process that jumps between discrete states, with the next state depending only on the present one. Its distribution $P_t(g, r) = \Pr\{G(t) = g,\, R(t) = r\}$ evolves according to the chemical master equation [13, 14, 15, 16]

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}P_t(g,r) = {} & k_{\text{on}}\big[\delta_{g,1}P_t(0,r) - \delta_{g,0}P_t(g,r)\big] \\ & + k_{\text{off}}\big[\delta_{g,0}P_t(1,r) - \delta_{g,1}P_t(g,r)\big] \\ & + k_{\text{syn}}\,\delta_{g,1}\big[P_t(1,r-1) - P_t(1,r)\big] \\ & + k_{\text{deg}}\big[(r+1)P_t(g,r+1) - r\,P_t(g,r)\big], \end{aligned} \tag{3}$$

with the convention $P_t(g, -1) = 0$. Equation (3) underlies all three approaches below: the SSA samples trajectories whose distribution solves it, the moment ODEs are derived from it exactly, and the steady-state sampler draws from its $t \to \infty$ fixed point.

3 Methods

We describe the three implementations in turn, then state in Section 3.4 the exact relationships that let them test one another.

3.1 Stochastic Simulation (Gillespie SSA)

The *stochastic simulation algorithm* (SSA) of Gillespie [1, 2, 3] generates statistically exact sample paths of a CTMC, making no approximation to the master equation (3). Its key idea is that, given the current state, two questions separate: when does the next reaction occur, and which reaction is it? With $a_0 = \sum_{j=1}^{4} a_j$ the *total reaction intensity*, the waiting time is $\tau \sim \text{Exp}(a_0)$ and reaction j fires with probability a_j/a_0.

Our implementation draws these by inversion, using two independent uniforms $r_1, r_2 \sim \text{U}(0, 1)$ per step:

$$\tau = \frac{1}{a_0}\,\ln\frac{1}{r_1}, \qquad j = \min\Big\{k : \textstyle\sum_{i=1}^{k} a_i > r_2\, a_0\Big\}. \tag{4}$$

The time is then advanced by τ, the state updated by reaction j, the reaction intensities recomputed, and the loop repeats for a fixed number of events n_{sim}. If the total reaction intensity reaches $a_0 = 0$, no reaction can fire, so the state is absorbing and is held fixed afterwards.

A single trajectory is the history of one cell. To obtain population statistics, we run N_{rep} independent trajectories with identical parameters and average over them. Each trajectory advances by its own random time increments, so the same event index occurs at different times in different trajectories. We therefore resample every trajectory onto a shared uniform time grid by a zero-order hold before averaging; the state is piecewise constant between events, so this introduces no error. The empirical moments are then ensemble averages at each grid time, for example

$$\widehat{\mu_R}(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t), \qquad \widehat{\mu_G}(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i(t), \tag{5}$$

with sample variances and the sample gene–RNA covariance defined analogously.

Random number generation. Because every result in this paper is an average over pseudo-random draws, we state the generator explicitly. All experiments use the Mersenne Twister MT19937 [20] through `numpy.random` [17]. Its period of $2^{19937} - 1$ is vast compared with the $\mathcal{O}(10^7)$ variates consumed by the largest run reported here, so exhausting the generator is not a concern. A seed is fixed for each experiment, so every figure in this paper is exactly reproducible from the accompanying code.

3.2 Moment ODEs

The SSA describes individual cells. Often we instead want deterministic equations for population-level statistics—the *moments*. A moment is an expectation: the mean $\mu_R = \mathbb{E}[R]$ is the average mRNA level, the variance $\sigma_R^2 = \mathbb{E}[R^2] - \mathbb{E}[R]^2$ measures the spread (the noise), and the covariance $C_{RG} = \mathbb{E}[RG] - \mathbb{E}[R]\mathbb{E}[G]$ measures how gene activity and mRNA level vary together.

Applying the standard moment rule for a chemical master equation—the expected rate of change of any observable is the sum over reactions of each intensity times the change it makes—to $f = G, R, R^2, GR$ in turn yields four coupled equations for the telegraph model [16, 15]:

$$\frac{\mathrm{d}\mu_G}{\mathrm{d}t} = k_{\text{on}}\,(1-\mu_G) - k_{\text{off}}\,\mu_G, \tag{6}$$

$$\frac{\mathrm{d}\mu_R}{\mathrm{d}t} = k_{\text{syn}}\,\mu_G - k_{\text{deg}}\,\mu_R, \tag{7}$$

$$\frac{\mathrm{d}\sigma_R^2}{\mathrm{d}t} = 2k_{\text{syn}}\,C_{RG} - 2k_{\text{deg}}\,\sigma_R^2 + k_{\text{syn}}\,\mu_G + k_{\text{deg}}\,\mu_R, \tag{8}$$

$$\frac{\mathrm{d}C_{RG}}{\mathrm{d}t} = k_{\text{syn}}\,\sigma_G^2 - (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})\,C_{RG}. \tag{9}$$

Each has a direct reading. Activation pushes mean activity toward 1 and inactivation pulls it toward 0; mRNA is made in proportion to the active fraction μ_G and removed in proportion to the amount present; the mRNA variance is driven up by the coupling to gene activity (the C_{RG} term) and by the randomness of production and decay, and down by degradation; and the covariance is created by transcription passing the gene's own fluctuations σ_G^2 on to the mRNA, then damped at the combined rate of switching and degradation.

Exact closure. In a general nonlinear reaction system the equation for each moment depends on higher moments, so the system never closes and one has to add approximate closure schemes [12, 16, 15]. Here it closes *exactly*. The only possible nonlinearity is the variance of the gene state, but because G is binary,

$$\sigma_G^2 = \mathbb{E}[G^2] - \mu_G^2 = \mathbb{E}[G] - \mu_G^2 = \mu_G\,(1-\mu_G), \tag{10}$$

which depends on the mean μ_G alone. Putting (10) into (9) turns the four equations (6)–(9) into a *closed, exact* linear system in $(\mu_G, \mu_R, \sigma_R^2, C_{RG})$. This is not an approximation: the solutions are the true moments of the master equation (3). This exact closure is the one property of the system we exploit directly: it is what makes a four-variable numerical integration sufficient, with no closure approximation anywhere in the code.

Implementation. We do not integrate the system by hand. The four right-hand sides (6)–(9), with σ_G^2 substituted by (10), are coded directly and passed to the explicit Runge–Kutta solver `scipy.integrate.solve_ivp` [18] (method `RK45`, the embedded Dormand–Prince pair [19]) from the initial conditions $\mu_G(t_0) = g_0$, $\mu_R(t_0) = r_0$, $\sigma_R^2(t_0) = C_{RG}(t_0) = 0$, evaluated on a uniform grid of 1000 time points.

3.3 Steady-State Sampler

As $t \to \infty$ the telegraph model forgets its initial condition and converges to a unique stationary distribution $P_\infty(g, r)$. Instead of simulating a long transient to reach it, we can sample from P_∞ directly. Peccoud and Ycart [4] showed that at steady state the mRNA count follows the classic *Beta–Poisson* law: it is Poisson, but with a mean that is itself random from cell to cell. Let the *mixing fraction* $p \in [0, 1]$ be the long-run share of time a cell's gene spends ON. Across a population p is Beta-distributed, and given p the mRNA count is Poisson with mean $\lambda = (k_{\text{syn}}/k_{\text{deg}})\, p$. Here $k_{\text{syn}}/k_{\text{deg}}$ is the *transcription efficiency coefficient*: the mean mRNA level a permanently active gene would reach. With the dimensionless switching rates $a = k_{\text{on}}/k_{\text{deg}}$ and $b = k_{\text{off}}/k_{\text{deg}}$ (so k_{deg} fixes the unit of time), the stationary mRNA law is therefore

$$R \sim \text{Poisson}\big((k_{\text{syn}}/k_{\text{deg}})\, p\big), \qquad p \sim \text{Beta}(a, b). \tag{11}$$

Our sampler implements exactly this law, and draws the gene state as an explicit Bernoulli(p) so we can also estimate the joint moments of (G, R). For each of the N_{rep} samples it draws

$$p \sim \text{Beta}(a, b), \qquad f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\,\Gamma(b)}\, p^{\,a-1}(1-p)^{\,b-1}, \tag{12}$$

$$G \mid p \sim \text{Bernoulli}(p), \qquad \Pr(G = 1 \mid p) = p, \quad \Pr(G = 0 \mid p) = 1 - p, \tag{13}$$

$$R \mid p \sim \text{Poisson}(\lambda), \qquad \lambda = \frac{k_{\text{syn}}}{k_{\text{deg}}}\, p. \tag{14}$$

The key point is that the mRNA count R is conditioned on the mixing fraction p, *not* on the current gene state G. Given p, the gene state and the mRNA count are drawn independently, so a cell can have $G = 0$ and still $R > 0$—as in a real cell, where mRNA made during the last active period outlives the switch to OFF.

This construction makes a sharp prediction at the edge of parameter space. As the switching rates become small compared with degradation $(a, b \to 0)$, the Beta(a, b) density piles its mass at the endpoints $p = 0$ and $p = 1$, and the mixture separates into its building blocks: a single Bernoulli trial for the gene state and, when ON, an ordinary Poisson count for the mRNA. In particular at $k_{\text{off}} = 0$ the mixing fraction is degenerate, $p \equiv 1$, and the stationary law must collapse to a plain Poisson$(k_{\text{syn}}/k_{\text{deg}})$. We test that prediction numerically in Section 4.3.

The joint stationary law is the average of these conditional distributions over the mixing fraction,

$$P_\infty(g, r) = \int_0^1 \Pr(G = g \mid p)\; \Pr(R = r \mid p)\; f(p)\,\mathrm{d}p, \tag{15}$$

which our sampler never evaluates directly: drawing N_{rep} triples (p, G, R) and averaging gives unbiased Monte Carlo estimates of the steady-state moments $\widehat{\mu_G}^{\,ss}, \widehat{\mu_R}^{\,ss}$, and so on. Two edge cases are handled explicitly. If $k_{\text{off}} = 0$ the gene never switches off, so p is fixed at 1 rather than drawn; if $k_{\text{deg}} = 0$ no steady state exists (mRNA grows without bound) and the input is rejected. The exact stationary means used as the reference below are

$$\mu_G^{ss} = \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \qquad \mu_R^{ss} = \frac{k_{\text{syn}}}{k_{\text{deg}}}\,\mu_G^{ss}, \tag{16}$$

obtained by setting the time derivatives in (6)–(7) to zero.

3.4 The Validation Triangle

The three methods are different mathematical objects, but probability theory predicts exact relationships between them. We state these as the three edges of a *validation triangle* and test each one numerically in the experiments that follow.

Edge 1 — SSA ↔ Moment ODEs (finite time). The empirical averages (5) are sample means of independent, identically distributed trajectories. By the Law of Large Numbers, as $N_{\text{rep}} \to \infty$ they converge to the true expectations, which are exactly the solutions of the moment ODEs:

$$\lim_{N_{\text{rep}}\to\infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i(t) = \mu_G(t), \qquad \lim_{N_{\text{rep}}\to\infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t) = \mu_R(t), \tag{17}$$

and likewise the empirical covariance converges to $C_{RG}(t)$. This links the stochastic and deterministic descriptions: with finitely many replicates, the SSA should follow the ODE curves up to a statistical error of order $1/\sqrt{N_{\text{rep}}}$.

Edge 2 — Moment ODEs ↔ Steady-state sampler (long time). The stationary distribution is the $t \to \infty$ limit of the dynamics, so the moment ODEs must approach a fixed point,

$$\lim_{t\to\infty} \mu_G(t) = \mu_G^{ss}, \qquad \lim_{t\to\infty} \mu_R(t) = \mu_R^{ss}, \qquad \text{etc.}, \tag{18}$$

which coincides with (16). The independent steady-state sampler must estimate *these same* values,

$$\lim_{N_{\text{rep}}\to\infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i^{ss} = \mu_G^{ss}, \qquad \lim_{N_{\text{rep}}\to\infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i^{ss} = \mu_R^{ss}. \tag{19}$$

By the Central Limit Theorem, the sampler's error decays as $\sigma/\sqrt{N_{\text{rep}}}$, where σ is the standard deviation of the stationary distribution. Observing this $1/\sqrt{N_{\text{rep}}}$ rate confirms that the sampler is a valid Monte Carlo estimator of the steady-state moments.

Edge 3 — SSA ↔ Steady-state sampler (closing the triangle). An SSA trajectory run long enough to reach stationarity samples the same distribution P_∞ as the direct sampler. Hence the long-time plateau of the SSA moments must agree with both the ODE fixed point and the sampler estimates.

4 Experiments and Results

The study is organised into three Jupyter notebooks, one for each part of the validation. Notebook 1 (`ssa_parameter_exploration.ipynb`) explores the SSA on its own to build intuition; Notebook 2 (`ssa_vs_ode.ipynb`) tests Edge 1 by comparing the SSA with the moment ODEs; Notebook 3 (`validation.ipynb`) tests Edges 2 and 3 by checking the steady-state sampler against closed-form results.

Unless stated otherwise, the dynamic experiments use the symmetric parameters $k_{\text{on}} = 0.5$, $k_{\text{off}} = 0.5$, $k_{\text{syn}} = 20.0$, $k_{\text{deg}} = 1.0$ (time measured in mRNA lifetimes), with $N_{\text{rep}} = 1000$ trajectories and initial condition $(g_0, r_0) = (0, 0)$. These give an active fraction $\mu_G^{ss} = k_{\text{on}}/(k_{\text{on}} + k_{\text{off}}) = 0.5$ and steady-state mean mRNA level $\mu_R^{ss} = (k_{\text{syn}}/k_{\text{deg}})\,\mu_G^{ss} = 10$, by (16). The five-moment sampler check in Section 4.3 uses $k_{\text{off}} = 1$ instead, which gives $\mu_R^{ss} = 6.667$.

4.1 Parameter Exploration (Notebook 1)

Setup. This part runs the SSA on its own and checks that it reproduces the behaviour we expect when the gene's parameters change. We vary one factor at a time so each figure isolates a single mechanism; the parameter values are listed in Table 1. To first see what a *single* cell does before averaging, Figure 1 shows one SSA trajectory: the gene switches ON and OFF, mRNA is made in bursts while it is ON, and each burst then decays away.

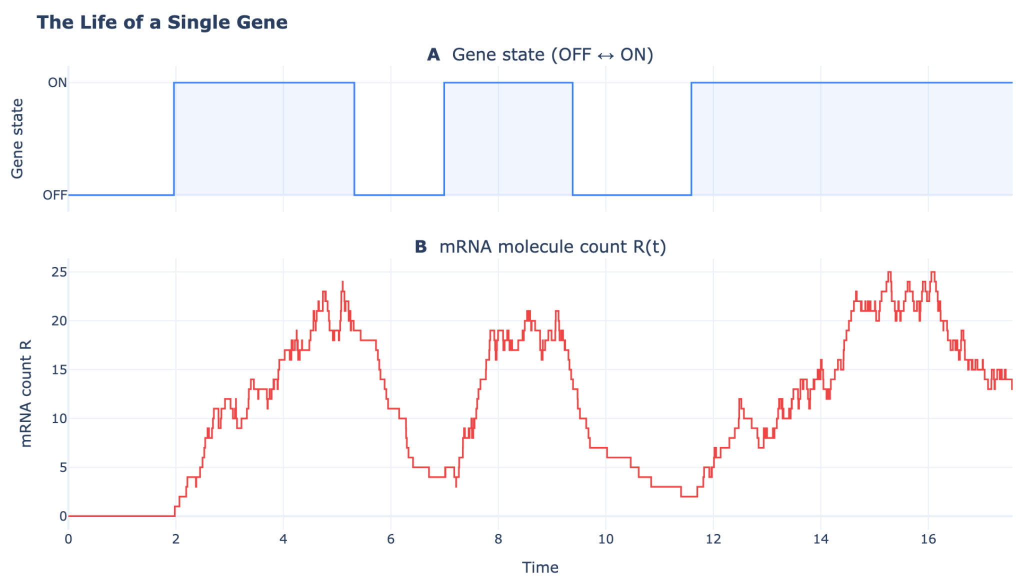

Figure 1: One stochastic run, i.e. a single sample path ("the life of a single gene"). Top: the gene state switches between OFF and ON at random times. Bottom: the mRNA count rises in bursts during ON periods and decays between them. Averaging many such trajectories gives the sample moments used below.

Figures. Switching speed (Figure 2) is set by how $k_{\text{on}}, k_{\text{off}}$ compare with k_{deg}; it reshapes the noise while the mean stays fixed, from long bursts and a wide band at slow switching to a narrow band at fast switching (details in the figure caption). Expression level (Figure 3) is set by the burst size $k_{\text{syn}}/k_{\text{off}}$ of (2) at fixed k_{on}: a strong promoter (burst $\approx$ 14) reaches a high mean, a weak

Experiment	Parameters	What it isolates
Slow switching	$k_{\text{on}} = k_{\text{off}} = 0.05,\ k_{\text{syn}} = 20$	noise at fixed mean
Fast switching	$k_{\text{on}} = k_{\text{off}} = 10,\ k_{\text{syn}} = 20$	noise at fixed mean
High expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 4, k_{\text{syn}} = 56$	burst size $k_{\text{syn}}/k_{\text{off}} \approx 14$
Low expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 10, k_{\text{syn}} = 30$	burst size $k_{\text{syn}}/k_{\text{off}} \approx 3$
Initial cond.	$(g_0, r_0) = (0, 0)$ vs. $(1, 15)$	forgetting the start
Numerical	$n_{\text{sim}} = 1000$ vs. $N_{\text{rep}} = 224,\ n_{\text{sim}} = 5000$	estimate quality

Table 1: The six parameter-exploration runs. Unless shown otherwise $k_{\text{deg}} = 1$ and the base switching rates are $k_{\text{on}} = k_{\text{off}} = 0.1$; each run reaches the steady-state mean $\mu_R^{ss} = 10$ except the expression runs, which change it on purpose.

promoter (burst ≈ 3) makes few molecules with large relative noise. Figure 4 runs the same rates from two very different starts, and Figure 5 varies only the numerical resolution.

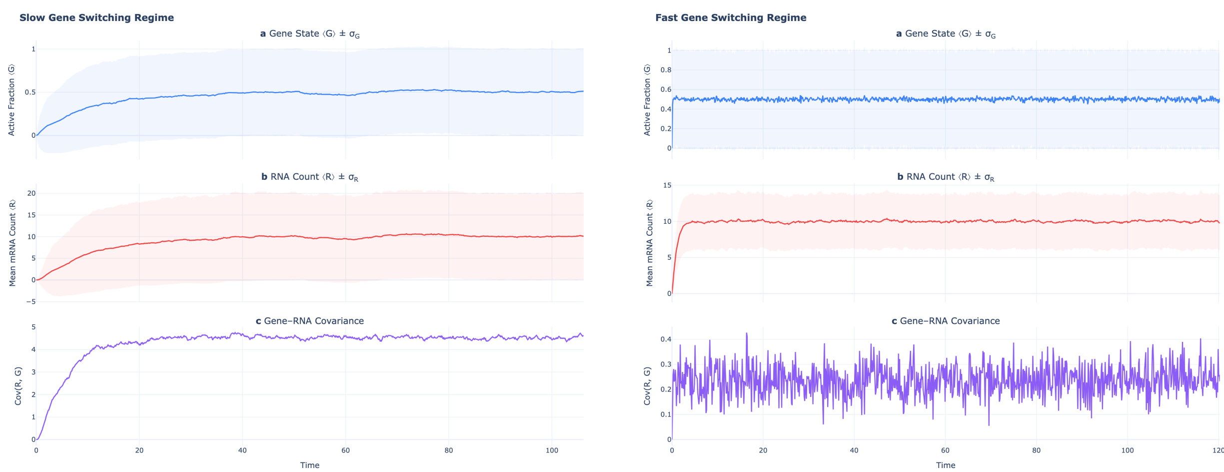

Figure 2: Switching speed reshapes noise at fixed mean. Slow switching (left, $k_{\text{on}} = k_{\text{off}} = 0.05$) gives long bursts, a wide $\pm\sigma_R$ band and strong gene–RNA covariance; fast switching (right, $k_{\text{on}} = k_{\text{off}} = 10$) flickers so quickly that the promoter averages out, leaving a narrow band and near-zero covariance. Both reach the same mean $\mu_R \approx 10$.

Analysis. Three mathematical points come out of these runs. First, switching speed reshapes the noise at a fixed mean: as $k_{\text{on}}, k_{\text{off}}$ grow relative to k_{deg} the gene averages out, so σ_R^2 and C_{RG} fall and the Fano factor $F = \sigma_R^2/\mu_R$ moves toward the Poisson baseline $F = 1$, exactly as the covariance equation (9) predicts and in agreement with the exact analysis of the switching limits in [11]. Second, the mean is governed by $\mu_R^{ss} = (k_{\text{syn}}/k_{\text{deg}})\, k_{\text{on}}/(k_{\text{on}} + k_{\text{off}})$, so with k_{on} fixed the burst size (2) sets how high μ_R climbs; low expression has a larger *relative* spread (coefficient of variation σ_R/μ_R), which is why rare-molecule genes look noisier. Third, both initial conditions relax to the same fixed point, confirming that the chain is ergodic (it forgets where it started). The numerical run then shows the estimate—not the biology—improving: the sampling error scales as $1/\sqrt{N_{\text{rep}}}$, with extra bias if runs are too short to reach stationarity. Here $N_{\text{rep}} = 224$ is a representative experiment-scale ensemble, of the order of the cell counts in single-molecule mRNA studies [5].

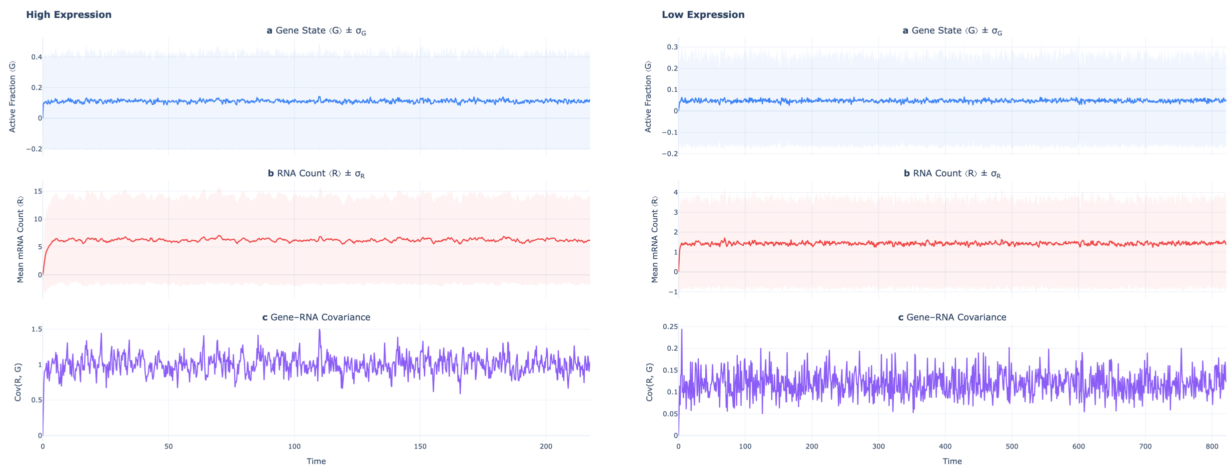

Figure 3: Expression level is set by burst size $k_{\mathrm{syn}}/k_{\mathrm{off}}$. A strong promoter (left, burst ≈ 14) reaches a high mean mRNA level; a weak promoter (right, burst ≈ 3) yields few molecules and large relative fluctuations.

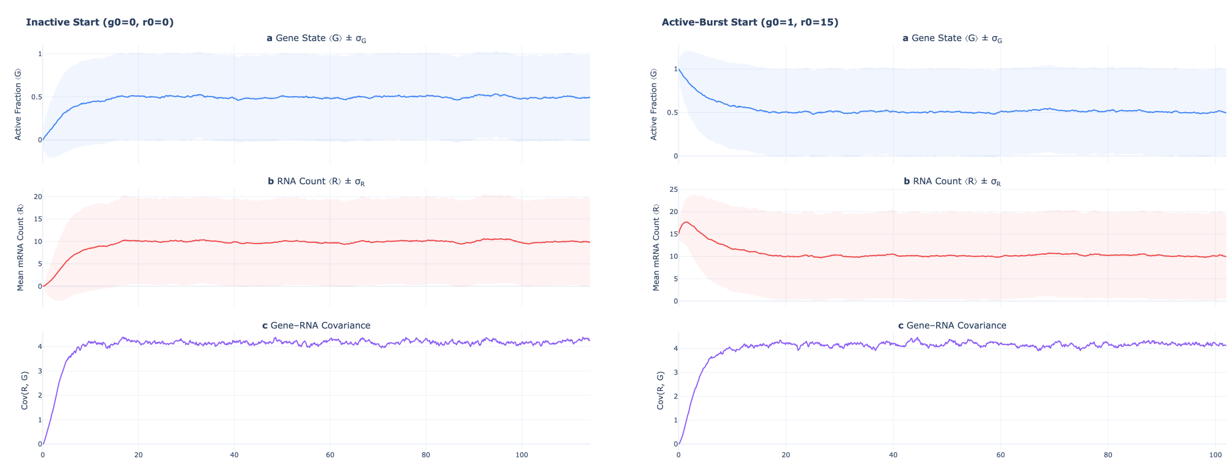

Figure 4: The steady state does not depend on initialisation. An inactive start (g_0=0, r_0=0, left) builds up from zero while an active-burst start (g_0=1, r_0=15, right) relaxes downward; both converge to the same stationary mean.

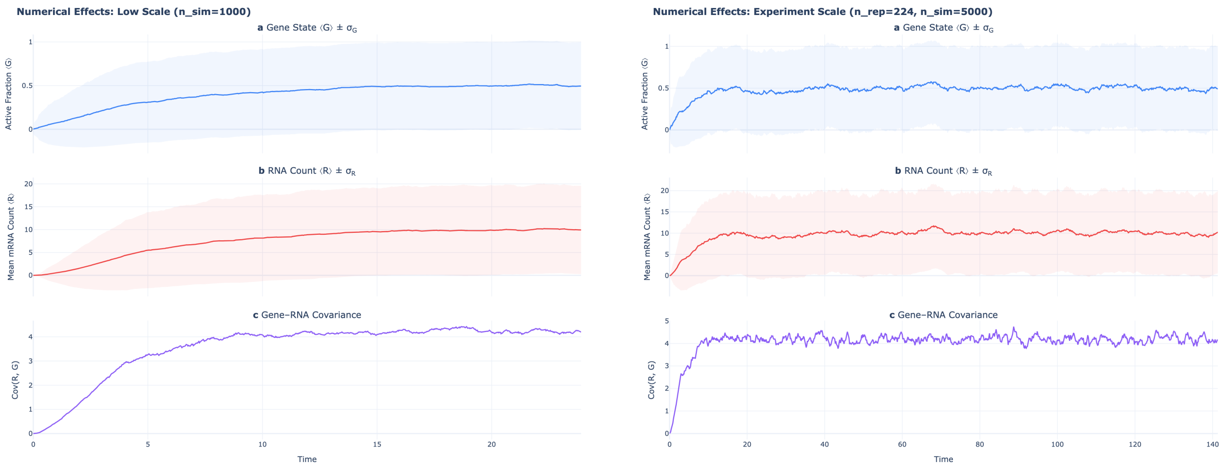

Figure 5: Numerical resolution controls estimate quality, not the biology. A short, under-sampled run (left, $n_{\text{sim}} = 1000$) gives jagged moments; the experiment-scale run (right, $N_{\text{rep}} = 224$, $n_{\text{sim}} = 5000$) reaches steady state with smooth estimates.

4.2 SSA vs. ODE: Moment Dynamics (Edge 1, Notebook 2)

Aim. Verify that the stochastic simulation and the deterministic moment ODEs describe the same dynamics at all times (Edge 1 of the validation triangle, Section 3.4). If they agree, the SSA ensemble averages should track the ODE curves up to sampling noise of order $1/\sqrt{N_{\text{rep}}}$.

Result. The direct test is the overlay in Figure 6, which shows both descriptions of the same experiment on one set of axes. Starting from a silent gene with no mRNA, the mean activity $\mu_G(t)$ relaxes to its stationary value 0.5, the mean mRNA count $\mu_R(t)$ rises toward 10, and the gene–RNA covariance $C_{RG}(t)$ rises and then plateaus. The SSA ensemble averages (green, with $\pm\sigma$ bands) lie on top of the deterministic ODE curves (dashed blue) in all three panels, throughout the transient and into the steady state. The ODE curves are smooth, being exact expectations rather than estimates; the only visible difference is a small wiggle in the SSA covariance curve, the expected finite-N_{rep} sampling fluctuation, which shrinks as more trajectories are added. The shaded bands are worth reading separately from the agreement: the spread in mRNA count across cells is large, of order the mean itself, and that is the phenotypic variability the telegraph model exists to capture.

The agreement is not limited to this symmetric case: Notebook 2 repeats the comparison across slow and fast switching, high and low expression, and both initial conditions, and the SSA means fall on the ODE curves in every regime. Figure 7 shows the hardest case, slow switching, where the noise is largest; the SSA still tracks the ODE throughout.

Conclusion. The SSA converges to the exact moment dynamics, as predicted by the Law of Large Numbers, so Edge 1 holds for any parameters. The $\pm\sigma$ bands do *not* collapse onto the mean: they measure genuine cell-to-cell variability, which persists no matter how many trajectories are averaged and is correctly predicted by the variance equation (8).

4.3 Steady-State Validation (Edges 2 & 3, Notebook 3)

Aim. Verify that the long-time limit of the moment ODEs agrees with the independent steady-state sampler (Edge 2), that the sampler's error decays at the Central Limit Theorem rate $1/\sqrt{N_{\text{rep}}}$, and that both reduce correctly to known limits at the boundaries of parameter space (Edge cases).

Result. The ODE solutions in Figure 6 plateau at $\mu_G^{ss} = 0.5$ and $\mu_R^{ss} = 10$, matching the

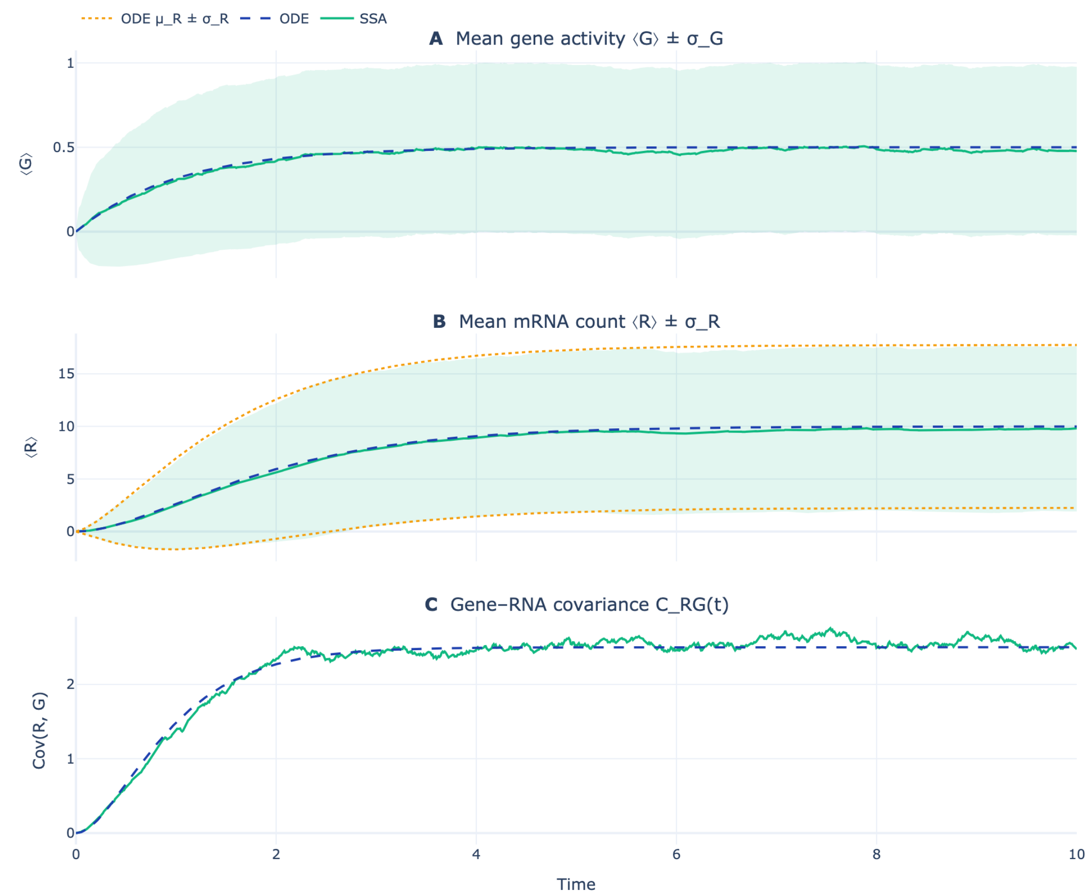

Figure 6: Comparison of the SSA ensemble averages (green, with the empirical $\pm\sigma$ fluctuation bands shaded) and the deterministic ODE solutions (dashed blue). In the mRNA panel the analytical spread from the moment ODEs, $\mu_R \pm \sigma_R$, is drawn as amber dotted lines on top of the SSA band, so the stochastic scatter can be checked directly against the exact variance dynamics. The SSA results closely match the ODE predictions, validating the stochastic simulation against the exact moment dynamics.

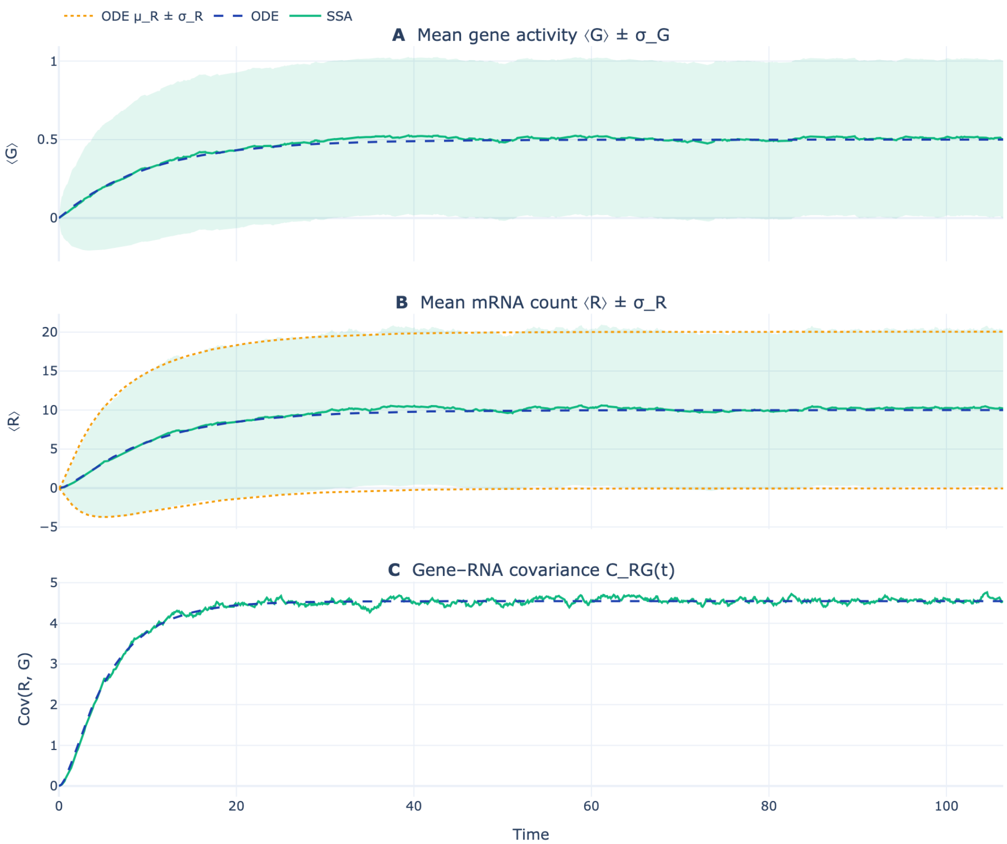

Figure 7: SSA versus moment ODEs in the high-noise slow-switching regime. Colours are as in Figure 6: green SSA means with shaded $\pm\sigma$ bands, dashed blue ODE means, and the amber dotted analytical $\mu_R \pm \sigma_R$ overlay in the mRNA panel. Even where fluctuations are largest, the SSA ensemble averages track the deterministic ODE curves throughout the transient and into steady state, confirming Edge 1 beyond the symmetric reference case.

analytical fixed point (16). The direct sampler reproduces these values: with $N_{\text{rep}} = 2 \times 10^5$ samples it returns $\widehat{\mu_G}^{ss} \approx 0.499$ and $\widehat{\mu_R}^{ss} \approx 10.02$, agreeing to within Monte Carlo error. Figure 8 makes the convergence quantitative: on log–log axes, the mean absolute deviation of the sampler's μ_R^{ss} estimate from the exact value follows a line of slope $-\frac{1}{2}$, matching the $1/\sqrt{N_{\text{rep}}}$ reference over four decades of sample size.

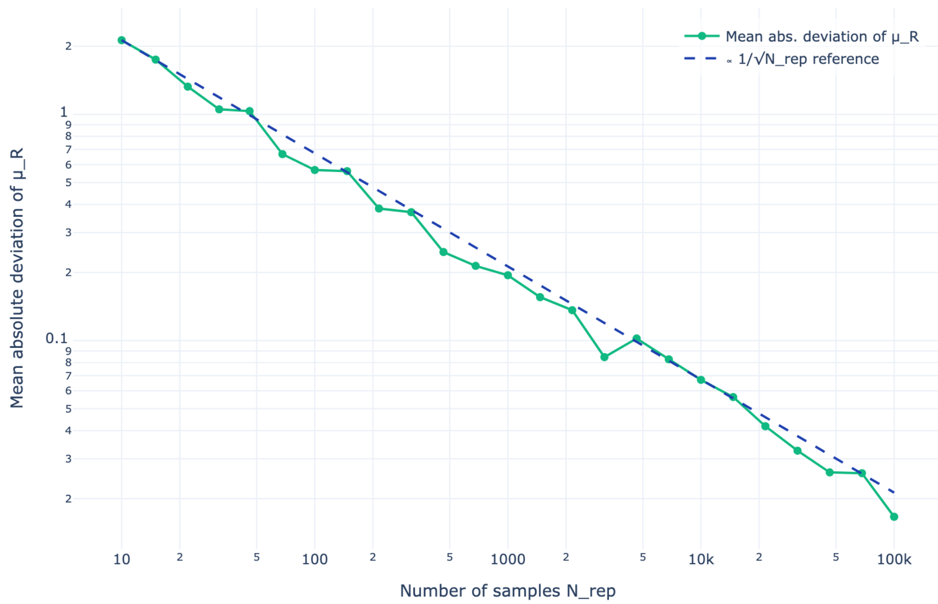

Figure 8: Convergence of the exact steady-state sampler's empirical moments to the analytical steady-state values as a function of the number of samples N_{rep}. The error (shown as mean absolute deviation) decays as $1/\sqrt{N_{\text{rep}}}$, as expected from the Central Limit Theorem.

A more demanding check asks the sampler to reproduce *all five* steady-state moments at once. For $k_{\text{on}} = 0.5, k_{\text{off}} = 1, k_{\text{syn}} = 20, k_{\text{deg}} = 1$ (so $\mu_R^{ss} = 6.667$), each moment has a closed form,

$$\mu_G^{ss} = \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \quad \mu_R^{ss} = \frac{k_{\text{syn}}}{k_{\text{deg}}}\mu_G^{ss}, \quad C_{RG}^{ss} = \frac{k_{\text{syn}}\,\mu_G^{ss}(1-\mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}, \tag{20}$$

together with $\sigma_G^{ss} = \sqrt{\mu_G^{ss}(1-\mu_G^{ss})}$ and $\sigma_R^{ss} = \sqrt{\mu_R^{ss} + (k_{\text{syn}}/k_{\text{deg}})\,C_{RG}^{ss}}$. At $N_{\text{rep}} = 2 \times 10^5$, the sampler matches every moment to within 0.04 (Table 2), and the absolute error in μ_R shows the same $1/\sqrt{N_{\text{rep}}}$ decay, falling from 3.87 at $N_{\text{rep}} = 10$ to 0.0145 at $N_{\text{rep}} = 10^5$.

Finally, we test the two limits where the model reduces to a known result, including the $k_{\text{off}} = 0$ prediction made in Section 3.3. When the gene never switches off, it is permanently active and the telegraph model reduces to a birth–death process—a Markov chain whose only moves are $+1$ and -1—whose stationary law is a Poisson distribution [13, 14]; the moment ODEs accordingly relax to $\mu_G^{ss} = 1$, $\mu_R^{ss} = k_{\text{syn}}/k_{\text{deg}}$. Figure 9 overlays the sampled counts on the analytic Poisson PMF; they coincide, with $\mu_R \approx 9.99$ and a Fano factor (the variance-to-mean ratio) ≈ 0.99, against the expected mean $k_{\text{syn}}/k_{\text{deg}} = 10$ and Fano factor 1. In the opposite limit, with no degradation ($k_{\text{deg}} = 0$), mRNA accumulates without bound and no steady state exists; the sampler detects this and raises a `ValueError` instead of returning an invalid result.

Conclusion. The sampler is an unbiased, $\sqrt{N}$-consistent estimator of the steady-state moments, so Edge 2 holds. Because the SSA moments (Figure 6) plateau at these same values, Edge 3 holds

Moment	Analytic	Sampler	Abs. diff.
μ_G^{ss}	0.3333	0.3317	0.0016
μ_R^{ss}	6.6667	6.6277	0.0389
σ_G^{ss}	0.4714	0.4708	0.0006
σ_R^{ss}	6.4979	6.4767	0.0212
C_{RG}^{ss}	1.7778	1.7666	0.0111

Table 2: Steady-state moments from the direct sampler ($N_{\text{rep}} = 2 \times 10^5$) against their closed-form values (20), for $k_{\text{on}} = 0.5, k_{\text{off}} = 1, k_{\text{syn}} = 20, k_{\text{deg}} = 1$.

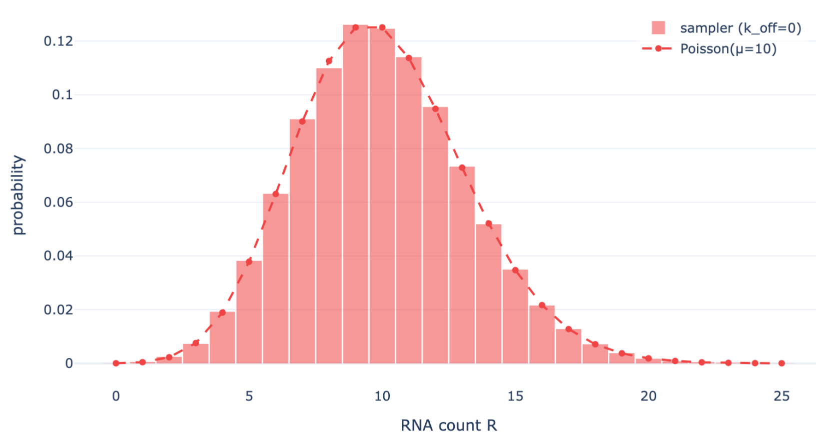

Figure 9: Edge case $k_{\text{off}} = 0$ (permanently active gene). The sampled steady-state mRNA distribution matches the analytic Poisson PMF with mean $k_{\text{syn}}/k_{\text{deg}} = 10$ (Fano factor ≈ 1), the exact limit the telegraph model must reduce to when the gene never switches off.

as well, and the three edges close the triangle, establishing mutual consistency of the stochastic, deterministic, and stationary descriptions. The correct behaviour at the $k_{\text{off}} = 0$ and $k_{\text{deg}} = 0$ boundaries shows the agreement is not an artifact of the particular parameter set.

5 Conclusion

We have studied the telegraph model of stochastic gene expression from three mathematically distinct viewpoints—exact stochastic simulation, closed moment ODEs, and direct steady-state sampling—and shown that they agree. Each way was confirmed numerically to within the expected $1/\sqrt{N_{\text{rep}}}$ statistical error, and the methods were further validated on solvable edge cases.

Beyond verifying three correct implementations, this work illustrates a general principle: when the same system can be described at several levels of abstraction, those descriptions become mutual tests.

6 Future Work

The telegraph model is a starting point, and two extensions would make it more realistic.

Adding the protein level. mRNA is only an intermediate: it is translated into protein, which is usually what the cell actually uses. A natural next step is a three-stage model, gene → mRNA → protein, with two extra reactions: translation of protein from mRNA at rate k_{trans}, and protein degradation at rate $k_{\text{deg},p}$. The same tools carry over directly—the SSA simply gains two reactions, and the moment ODEs gain equations for the mean and variance of the protein count and its covariance with mRNA—so we could follow how noise is passed from the gene down to the protein, the level at which most measurements are made.

Networks of interacting genes. Real genes do not act alone: the protein made by one gene can switch another gene ON or OFF, forming a *gene regulatory network*. A compact way to describe such a network is to treat each gene's ON/OFF state as a spin in an *Ising model*, where ON and OFF play the role of the two spin states and the regulatory links act as couplings between spins. The obstacle is dimensionality: with n genes the joint distribution has 2^n gene configurations (times all the mRNA and protein counts), far too many to store or integrate directly. Here *machine-learning* methods—for example neural-network density estimators or variational approximations—give a practical way to approximate these high-dimensional probability distributions, learning a compact model of the stationary state instead of enumerating it. Combining the exact small-system checks of this paper with such approximate large-system methods is a promising direction.

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